

Option Pricing with Bachelier Model

Ian Castleton

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1 Call option

$$F_T = F_t + \sigma W_T \quad (1)$$

$$= F_t + \sigma N(0, T) \quad (2)$$

$$= F_t + \sigma \sqrt{T} N(0, 1) \quad (3)$$

Substitute the variable y which is a sample from the standard normal distribution, $y = N(0, 1)$. We then have

$$F_T = F_t + \sigma \sqrt{T} y. \quad (4)$$

We note that F_T is already martingale i.e. it has zero drift. The price of the call option is the discounted payoff under the measure Q , namely

$$c = \mathbb{E}_Q \left[(F_T - K)_+ \right] \quad (5)$$

$$= \mathbb{E}_Q \left[(F_t + \sigma \sqrt{T} y - K)_+ \right] \quad (6)$$

$$= \mathbb{E}_Q \left[(F_t - K + \sigma \sqrt{T} y)_+ \right] \quad (7)$$

$$= e^{-r(T-t)} \int_{-\infty}^{\infty} (F_t - K + \sigma \sqrt{T} y)_+ dy \quad (8)$$

$$= e^{-r(T-t)} \int_{y_L}^{\infty} (F_t - K + \sigma \sqrt{T} y) \phi_y dy \quad (9)$$

where the lower limit y_L is found at the limit where the terms in parenthesis $(\dots)_+$ reach zero, namely

$$F_t - K + \sigma \sqrt{T} y_L = 0 \quad (10)$$

$$\sigma \sqrt{T} y_L = K - F_t \quad (11)$$

$$y_L = \frac{K - F_t}{\sigma \sqrt{T}}. \quad (12)$$

and ϕ_y is the density of variable y , namely the probability density function $N'(y)$.

Multiplying out the terms, the call option integral can be written in two parts,

$$c = \underbrace{e^{-r(T-t)} \int_{y_L}^{\infty} (F_t - K) \phi_y dy}_{I_1} + \underbrace{e^{-r(T-t)} \int_{y_L}^{\infty} \sigma \sqrt{T} y \phi_y dy}_{I_2} \quad (13)$$

Working on the first integral we have

$$I_1 = e^{-r(T-t)} \int_{y_L}^{\infty} (F_t - K) \phi_y dy \quad (14)$$

$$= e^{-r(T-t)} (F_t - K) \int_{y_L}^{\infty} \phi_y dy \quad (15)$$

$$= e^{-r(T-t)} (F_t - K) \int_{-\infty}^{-y_L} \phi_y dy \quad (16)$$

$$= e^{-r(T-t)} (F_t - K) \int_{-\infty}^{-y_L} N'(y) dy \quad (17)$$

$$= e^{-r(T-t)} (F_t - K) N(-y_L) \quad (18)$$

$$= e^{-r(T-t)} (F_t - K) N(d) \quad (19)$$

where

$$d = \frac{F_t - K}{\sigma \sqrt{T}}. \quad (20)$$

Now looking at the second integral,

$$I_2 = e^{-r(T-t)} \int_{y_L}^{\infty} \sigma \sqrt{T} y \phi_y dy \quad (21)$$

$$= e^{-r(T-t)} \left[-\sigma \sqrt{T} \phi(y) \right]_{y_L}^{\infty} \quad (22)$$

$$= e^{-r(T-t)} \left[-\sigma \sqrt{T} N'(y) \right]_{y_L}^{\infty} \quad (23)$$

$$= e^{-r(T-t)} \sigma \sqrt{T} N'(y_L) \quad (24)$$

$$= e^{-r(T-t)} \sigma \sqrt{T} N'(d) \quad (25)$$

since $N'(x)$ is a symmetric function. Putting the two pieces together yields the final call option price

$$c = e^{-r(T-t)} \left[(F_t - K) N(d) + \sigma \sqrt{T} N'(d) \right]. \quad (26)$$